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Operator-Theoretic Decomposition of Market Fragility: The Clock of Regimes Model*

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Abstract:

We applied the Clock of Regimes (COR) model to analyze financial fragility through survival dynamics and ruin probability for the E-mini S&P 500 futures (ES) and S&P 500 index (SPX) during the 2007–2009 Global Financial Crisis. By rejecting ergodicity, the study embeds heavy-tailed regime switching within an open architecture where probability mass dissipates through regime-specific hazard rates. The COR model employs the operator $N = -K^{-1}$, extending classical Hidden Markov Models (HMMs) by introducing an exit parameter ϵ , transforming the system into a transient open structure. This yields the fundamental matrix N , which integrates expected state occupancy, making regime duration and exposure structurally observable. Fragility decomposes into two channels: duration fragility, where markets are trapped in adverse conditions, and catastrophe fragility, involving severe tail losses. Our empirical results show that Student- t specifications outperform Gaussian alternatives in capturing discontinuous shocks. Findings reveal distinct failure pathways: SPX exhibits stronger duration fragility through prolonged entrapment (132.14 days), whereas ES demonstrates stronger catastrophe fragility, with a ruin probability 1.6 times higher over 250 days. The COR framework provides a tractable operator-based approach for quantifying structural ruin and systemic financial fragility.

Keywords: Fat tails, Hazard, Survival, Residence Time, Markov Switching Models.

1 Introduction

We applied the Clock of Regimes (COR) model, a novel duration-aware stochastic framework that extends classical Hidden Markov Models (HMM), by embedding survival and hazard calculus directly into regime-switching dynamics $???$. By shifting the focus from one-step transition probabilities $??$ to a continuous-time operator $N = -K^{-1} ??????????$, we transform latent market states into a measurable temporal geometry of probability mass¹.

Within a Talebian framework, *fragility* is excessive concentration of probability mass in adverse states over time $?$. The COR model formalizes *antifragility* by contracting exposure as hazard intensifies and redeploying convexity as stress dissipates. Tail probabilities become state-conditional objects weighted by survival structure instead of unconditional frequencies.

Our study aimed to analyze financial fragility through the lens of survival dynamics and ruin probability of the E-mini S&P 500 futures (ES) and the S&P 500 index (SPX) during the 2007-2009 Global Financial Crisis crash. Our study demonstrates that a Student- t Hidden Markov Model (HMM) specification is statistically superior to Gaussian alternatives, uncovering a *path-dependent* market structure where steady regimes harbor hidden fragility and stress regimes act as high-intensity exit portals. The architecture culminates in a spectral fragility measure, which quantifies systemic risk not as a static probability, but as a structural *leak* of survival mass, ultimately providing a rigorous mathematical foundation for ruin-sensitive decision-making in nonlinear markets.

Our work contributes to the GFC literature by showing that crisis fragility is not one-dimensional. Using the COR survival operator $N = -K^{-1}$, we demonstrate that SPX exhibited duration fragility through prolonged crisis residence, whereas ES exhibited catastrophe fragility through higher tail intensity and a larger ruin bound. Thus, the GFC is decomposed into distinct survival pathways: entrapment versus explosive tail leakage.

2 Materials and Methods

2.1 The Clock of Regimes Model

Figure 1 illustrates the state-transition network of the Clock of Regimes (COR) model architecture by representing the three-state system corresponding to distinct structural market conditions. State 1 represents the **Stress Regime**, typically associated with periods of elevated volatility, liquidity

¹We define probability mass as the content of probability accumulated over an amount of time. In our COR architecture, probability is not treated purely as an instantaneous quantity $P(X)$. Instead, it is temporally embedded. It reflects how much probability resides in a state across time.

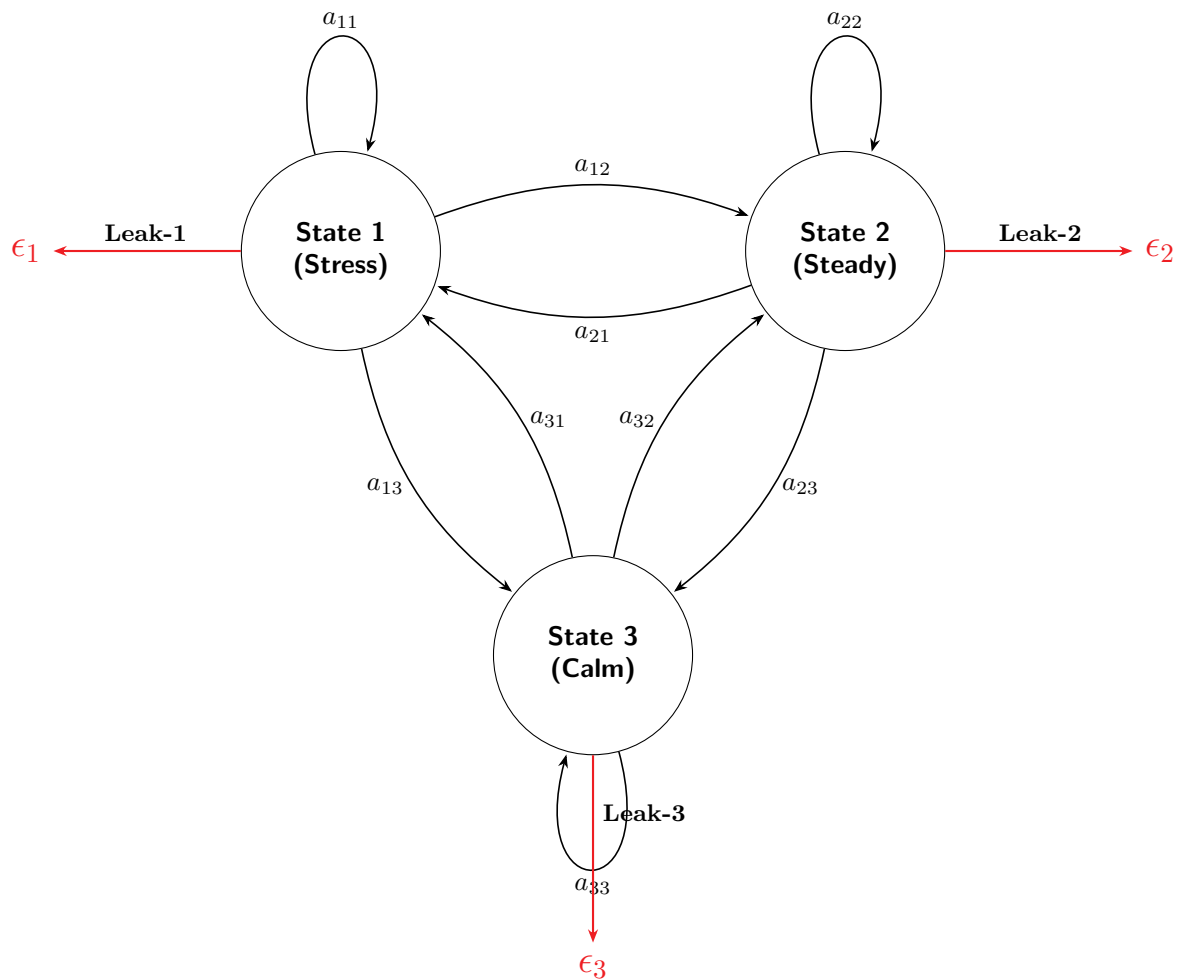


Figure 1: The Clock of Regimes model architecture.

deterioration, market panic, or systemic dislocation such as flash crashes and financial crises. State 2 corresponds to the **Steady Regime**, which captures standard transitional market behavior characterized by moderate volatility and ordinary market functioning. State 3 represents the **Calm Regime**, associated with low volatility, stable liquidity conditions, and equilibrium-like market behavior. The directional transitions between the states describe the movement of probability mass across regimes through time, thereby illustrating how markets evolve dynamically between calm, transitional, and crisis conditions within the COR survival-based framework. Unlike a standard, closed Hidden Markov Model (HMM) where probability mass is perfectly conserved inside the network, the COR architecture is explicitly designed as an open system with leakage (ruin parameters).

The black directed arrows show the standard, continuous-time intensities or probabilities (a_{ij}) of shifting from one active state to another. Self-loops (a_{11}, a_{22}, a_{33}) represent the persistence intensity associated with remaining within a particular regime through time.

Transition Paths ($a_{12}, a_{21}, a_{13}, a_{31}, a_{23}, a_{32}$) represent the dynamic bidirectional flow of probability mass between market regimes, capturing the evolving feedback structure of the financial system. These transition channels encode how markets migrate between calm, transitional, and stressed environments through time. For example, the transition from State 2 (*Steady*) to State 1 (*Stress*) reflects the escalation of systemic instability and volatility, whereas the reverse transition characterizes stabilization, recovery, and the re-establishment of market equilibrium.

The red outward-facing arrows represent the central innovation of the COR framework: regime-specific leakage channels (Leak-1, Leak-2, Leak-3), formally governed by the ruin intensities ϵ_1 , ϵ_2 , and ϵ_3 . These leakage operators transform the system from a closed Markov structure into an open survival architecture in which probability mass can dissipate over time through regime-dependent hazard mechanisms.

In this model (Figure 1), probability mass is actively allowed to exit the active state space from any node. This structural exit does not lead to another active trading regime, but rather out of the network entirely. This leakage mechanism directly models real-world system termination and breakdown events, including financial ruin and portfolio default. Within the COR framework, these leakage channels represent the dissipation of probability mass from the regime structure, thereby transforming the market from a closed persistence process into an open survival system subject to hazard, fragility, and collapse dynamics.

2.1.1 System Implications

Because of the active leakage parameters (ϵ_i), the row sums of the corresponding transition operator matrix Q (or sub-generator K) sum to strictly less than one (or less than zero in continuous-time

intensities):

$$\sum_j Q_{ij} = 1 - \epsilon_i < 1$$

This architectural layout transforms the transition matrix into a defective sub-Markovian generator. By allowing probability mass to escape through these red channels, the COR framework makes the expected duration, survival geometry, and exposure limits of the market completely observable.

2.1.2 Fragility Decomposition

The decomposition of fragility bridges the gap between observable, micro-level market shocks and the abstract, macro-level survival mathematics of the COR model. The residence-weighted tail exposure is defined as,

$$\bar{q} = \sum_{i=1}^K \pi_i q_i.$$

and, the average hazard leakage as,

$$\bar{\lambda} = \sum_{i=1}^K \pi_i \epsilon_i.$$

Component Mechanics: q_i , the conditional left-tail probability), represents the immediate, local risk of a severe negative shock occurring while the market is actively inside regime i . Because the COR model utilizes a heavy-tailed Student- t likelihood, it ensures that q_i is accurately calculated during crisis regimes, preventing the model from underestimating tail thickness (ν).

π , the survival-weighted regime distribution, is not a standard stationary distribution. It is derived from the fundamental matrix $N = (-K)^{-1}$. It represents the proportion of total remaining lifetime the system is expected to spend in each regime before leaking out into absolute ruin. It weights the states by their survival persistence.

The Aggregated Metrics: $\bar{q}$, the residence-weighted tail exposure), combines local tail risk with systemic duration. A market might have a terrifyingly high tail risk (q_i) in a specific crisis state, but if the system violently ejects from that state rapidly (low residence time π_i), the overall exposure $\bar{q}$ remains managed. Conversely, if the market gets trapped in a moderately risky state for a long time, $\bar{q}$ expands dramatically.

$\bar{\lambda}$, the average hazard leakage, aggregates the state-specific leakage/ruin rates (ϵ_i) across the expected lifetime of the system. It represents the global velocity at which probability mass is actively escaping the active network into system termination (default, liquidity collapse, ruin) over time.

The Central Intuition: The Dual Paths to Fragility A crucial concept regarding fragility and ruin ? is that a system can fail via two completely different pathways, and this decomposition structurally separates them:

$$\text{Systemic Risk} \propto f\left(\underbrace{\pi}_{\text{Persistence Architecture}}, \underbrace{q_i, \epsilon_i}_{\text{Shock Severity}}\right)$$

Pathway A: Regime Trapping (The Clock Component): The market’s structural transition matrix keeps it locked in an adverse state for an extended period (high π_i for a bad state). Even if individual daily shocks (q_i) are mild, the accumulation of time spent in a fragile environment triggers eventual collapse.

Pathway B: Rapid Ruin Accumulation (The Tail Component): The market’s structural transition matrix keeps it locked in an adverse state for an extended period (high π_i for a bad state). Even if individual daily shocks (q_i) are mild, the accumulation of time spent in a fragile environment triggers eventual collapse.

Summary This decomposition is where our mathematical operators (N and K) directly validate the COR model’s economic narrative. It proves that systemic risk is a multiplicative function of where the market sits in time (π) and how hard it gets hit when it is there (q_i, ϵ_i). It turns “market fragility” from a vague description into a crisp, operator-level metric.

2.1.3 Spectral Fragility

Let $\rho(N)$ denote the spectral radius of the fundamental matrix. An increase in $\rho(N)$ indicates expansion of survival-weighted residence time across regimes. In a perfectly safe, stationary system, the spectral radius equals exactly 1.0. 100% of survival mass is preserved daily. The COR model forces hazard leakage forces, ρ , to be strictly below 1.0. This small fractional deficit is the mathematical definition of systemic fragility.

By tracking the spectral radius $\rho(N)$ of the fundamental matrix on rolling data windows, the COR model acts as an early warning system of risk.

If *Fragility Decomposition* focused on the localized, state-by-state avenues of risk, *Spectral Fragility* provides a singular, global macro-metric for the entire system’s vulnerability using operator theory.

The Operator Mechanics of $\rho(N)$: The fundamental matrix $N = (-K)^{-1}$ acts as a temporal map, recording the expected total residence time the system will spend in every active state before

escaping. The spectral radius, denoted as $\rho(N)$, represents the largest absolute eigenvalue of this matrix:

$$\rho(N) = \max\{|\lambda_1|, |\lambda_2|, \dots, |\lambda_K|\}$$

In linear algebra, the spectral radius dictates the long-term growth or decay rate of an operator when it is applied repeatedly.

The Baseline: The “Perfectly Safe” Benchmark: The “Perfectly Safe” theoretical baseline is a completely closed, risk-free system where $\rho(N) = 1.0$. In a standard, classic Markov chain with no absorption states, probability mass circulates endlessly between states. No matter how long the clock ticks, 100% of the probability mass remains within the system.

The Fractional Deficit as Fragility: The central innovation here is how the COR model intentionally shatters this baseline. Because the sub-generator matrices (K or Q) incorporate active red leakage channels (ϵ_i), probability mass is continuously evaporating.

This forces the spectral radius of the fundamental matrix to drop below unity:

$$\rho(N) < 1.0$$

The distance between the actual spectral radius and the safe benchmark—the fractional deficit ($1.0 - \rho(N)$)—is not a mathematical error; it is the realization of systemic fragility. The lower $\rho(N)$ drops, the faster probability mass is bleeding out of the active state network, signaling that the entire market system is rapidly sliding toward structural ruin.

Application: The Rolling Early Warning System: By transitioning this concept from a static formula into an empirical tool, the COR model allows creation of a powerful market tracking mechanism. Calculating $\rho(N)$ on a rolling window over time series data, allows monitoring the structural health of a market in real-time.

Summary The spectral radius $\rho(N)$ elevates the COR model from a descriptive classifier into a definitive predictive risk diagnostic. It proves that systemic fragility doesn’t require you to guess when a *black swan* event will occur; instead, you can mathematically measure the structural degradation of the entire system’s matrix architecture by tracking a single, elegant eigenvalue metric.

2.1.4 Residence-Weighted Ruin Bound

Let T_h denote a risk horizon. Define the structural ruin probability bound

$$P(\text{ruin}) \leq 1 - \exp \left(-T_h \frac{\pi^\top N q}{\pi^\top N \mathbf{1}} \right).$$

This bound captures the interaction between regime duration, tail exposure, and hazard structure.

The *Residence-Weighted Ruin Bound* serves as the ultimate analytical capstone of our works's theoretical framework. It synthesizes all previous mathematical operators (N , q , and π) into a single, highly actionable probability metric: an upper bound on total systemic ruin over a specific forward-looking time horizon (T_h).

The Operator Mechanics of the Bound: The core mathematical engine of this inequality is the ratio embedded inside the exponential function:

$$\frac{\pi^\top N q}{\pi^\top N \mathbf{1}}$$

The denominator ($\pi^\top N \mathbf{1}$), the vector $\mathbf{1}$, acts as a summation operator. Multiplying the fundamental matrix N by $\mathbf{1}$ yields the total expected residence time across all combined active regimes. Weighting this by the survival-weighted initial state distribution ($\pi^\top$) gives the global expected life expectancy of the entire system before mass escapes through the leakage channels.

The Numerator ($\pi^\top N q$), instead of summing purely across time, maps the fundamental matrix against the heavy-tailed tail-risk vector q . This isolates the total expected exposure to extreme, left-tail shocks over the system's remaining lifetime.

The Time Horizon and the Exponential Boundary: By scaling this risk intensity by a specific terminal forward horizon (T_h), you wrap the expression in a classic Markovian survival probability structure ($1 - \exp(-\lambda t)$):

$$P(\text{ruin}) \leq 1 - \exp(-T_h \cdot \text{Risk Intensity})$$

This provides an incredibly elegant, non-linear bound on systemic failure. As the risk horizon T_h expands, or as the tail exposure in the numerator increases relative to the system's life expectancy, the negative exponent grows larger, driving the exponential term toward zero and forcing the $P(\text{ruin})$ bound asymptotically toward 1.0 (absolute certainty of ruin).

Economic and Philosophical Context (The Talebian Connection): This formulation mathematically realizes a core principle from Nassim Taleb’s *Incerto* ? : If a system has an open leakage channel or a non-zero probability of terminal ruin, it cannot be analyzed using standard stationary time-series statistics. Over a long enough horizon ($T_h \rightarrow \infty$), path dependency guarantees that the system will eventually hit a catastrophic transition or liquidity evaporation state. This formula allows a portfolio manager, risk officer, or strategist to explicitly input a specific trading horizon (e.g., $T_h = 30$ days or 90 days) and calculate the exact mathematical boundary of that ruin path.

3 Empirical Validation

This section describes the empirical implementation of the COR model. The methodology combines heavy-tailed HMM estimation with survival-adjusted operator construction in order to quantify systemic fragility in the financial return series of the E-mini S&P 500 futures (ES) and the S&P 500 index (SPX).

3.1 Data Management

The daily historical data for analysis was sourced via Yahoo Finance ?, which provides a robust and accessible foundation of historical data for both the ES (E-mini S&P 500 Futures) and the SPX (S&P 500 Index) time series. By utilizing closing prices, the model ensures that the calibration of daily paths remain anchored to widely recognized market benchmarks. This data allows for the precise mapping of regime transitions and the calculation of probability mass across the 2007–2009 epoch, capturing the high-fidelity volatility signatures required for the Clock of Regimes (COR) model.

The E-mini S&P 500 (ES) is a highly liquid futures contract traded on the Chicago Mercantile Exchange (CME) ?. ES is a derivative of the underlying S&P 500 index. Represented by tickers such as ES or ES=F, it allows traders to control a significant notional value—calculated as \$50 times the current index level—through the use of margin-based leverage.

The S&P 500 Index (SPX) is a market-capitalization-weighted index of 500 of the largest publicly traded companies in the United States ?. Represented by tickers such as SPX or ^GSPC, SPX is a theoretical benchmark—a cash index with a \$100 multiplier per index point—that cannot be traded directly; instead, investors gain exposure to it through exchange-traded funds (ETFs) like the SPY or through settled option contracts. CBOE SPX options are European-style, cash-settled contracts.

The COR analysis was conducted on historical daily return data. Let $\{P_t\}_{t=0}^T$ denote an observed

ES or SPX asset price series sampled at a fixed frequency. Log returns are defined as,

$$X_t = \log P_t - \log P_{t-1}, \quad t = 1, \dots, T.$$

All datasets used in the empirical comparison are aligned in sample length to ensure structural comparability of regime estimates and operator-based risk metrics.

4 Results

To capture the shifting market dynamics, the COR model identified three distinct hidden regimes (Stress, Steady, and Calm). In the COR model, these states are not just descriptive labels; they are formalized mathematical environments defined by the interaction of drift (μ), volatility (σ), and tail thickness (ν) characterized by unique volatility and drift parameters. The tail risk is formally evaluated at a probability level of $\alpha = 0.01$, allowing the model to focus on extreme left-tail events that represent potential systemic failures. Finally, these structural parameters are integrated over a ruin horizon of $T = 250$ trading days, providing a standardized one-year window to quantify the cumulative probability of wealth destruction or regime exit.

4.1 Model Specification

For both ES and SPX, the structural evaluation of the COR model analysis involves a formal comparison between two competing model specifications: a standard Gaussian Hidden Markov Model (HMM) and a Student- t Hidden Markov Model (t -HMM). While the Gaussian approach assumes normally distributed shocks within each regime, the Student- t specification explicitly accounts for the excess kurtosis often observed in ES and SPX returns during periods of systemic instability.

4.2 Findings for SPX

Empirically, the Student- t model demonstrates a statistically superior fit to the crash data, achieving a higher log-likelihood of $\ell_{\text{Student}} = 2147.54$ compared to $\ell_{\text{Gaussian}} = 2140.54$ (Table 1). The substantial gain in the log-likelihood (+7.00) confirms that a heavy-tailed distribution is necessary to accurately capture the discontinuous price movements and *flash* volatility that define the catastrophe fragility channel.

To further quantify model performance, we evaluated the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). These metrics provide a penalized assessment of the log-likelihoods, balancing the goodness-of-fit against the structural complexity of the model. As tabulated in Table 1, while the BIC slightly favors the Gaussian model due to its stricter penalty for

Table 1: Model Selection Criteria: Gaussian vs. Student- t HMM

Metric	Gaussian HMM ($k = 12$)	Student- t HMM ($k = 15$)	Decision
Log-Likelihood (ℓ)	2140.54	2147.54	<i>Student-t (+7.00)</i>
AIC	-4257.08	-4265.08	<i>Student-t (Lower is better)</i>
BIC	-4201.57	-4195.69	<i>Gaussian (Penalty for k)</i>

the three extra degrees of freedom ($k = 15$), the AIC and Log-Likelihood confirm that the Student- t model provides a fundamentally better representation of the heavy tails in the ES and SPX return series.

4.2.1 Transition Dynamics

Gaussian Transition Matrix A :

The Gaussian model identifies the following transition probabilities between the three identified regimes:

$$A_{\text{Gaussian}} = \begin{bmatrix} 0.9609 & 0.0391 & 0.0000 \\ 0.0110 & 0.9865 & 0.0024 \\ 0.0000 & 0.0074 & 0.9926 \end{bmatrix} \quad (1)$$

Student- t Transition Matrix A : The Student- t model, which provided a higher log-likelihood of 2147.54, identifies these switching dynamics:

$$A_{\text{Student-}t} = \begin{bmatrix} 0.9787 & 0.0213 & 0.0000 \\ 0.0159 & 0.9808 & 0.0033 \\ 0.0000 & 0.0076 & 0.9924 \end{bmatrix} \quad (2)$$

In both models, the diagonal elements are all > 0.95 , indicating high regime persistence where states are likely to remain the same from one day to the next. Both matrices show a 0.000000 probability of a direct jump between State 1 (Bull/Low Vol) and State 3 (Crisis/High Vol), suggesting the system must pass through the *neutral* State 2 to transition between extremes. The Student- t State 1 (0.9787) is more stable than the Gaussian State 1 (0.9608), which influences the subsequent COR generator K and the survival geometry of the model.

4.2.2 Hazard-Adjusted Survival Operator

The Hazard-adjusted survival operator Q serves as the core survival geometry within the COR model. It modifies the standard transition dynamics by accounting for the probability of emphescaping or

failing to survive a regime.

$$Q = \begin{bmatrix} 0.9567 & 0.0208 & 0.0000 \\ 0.0157 & 0.9705 & 0.0033 \\ 0.0000 & 0.0075 & 0.9770 \end{bmatrix} \quad (3)$$

Each row sum satisfies

$$\sum_j q_{ij} = 1 - \epsilon_i.$$

The matrix maintains a structural zero between State 1 and State 3, indicating that direct transitions between the extreme bull and crisis regimes are statistically impossible without passing through the intermediate State 2. Note that the row sums of Q are strictly < 1 .

4.2.3 The COR Generator Matrix

The COR generator K matrix is derived directly from the hazard-adjusted transition operator Q by the relation $K = Q - I$. This matrix represents the instantaneous rates of flow and *leakage* (risk) or hazard rate of each system state.

$$K = Q - I = \begin{bmatrix} -0.0433 & 0.0208 & 0.0000 \\ 0.0157 & -0.0295 & 0.0033 \\ 0.0000 & 0.0075 & -0.0230 \end{bmatrix}. \quad (4)$$

The diagonal elements of K represent the total exit rate from each state. State 1 has the highest exit rate at -0.043292 , indicating it is the most transient regime in this dataset. These values represent the rates of moving specifically from one regime to another. For example, the rate of moving from State 2 to State 1 is 0.015740 . Unlike a standard Markov generator where rows sum to zero, these rows sum to a negative value. This *missing* probability mass is the local hazard rate, which the model uses to calculate the Residence-weighted hazard ($\bar{\lambda}$) of 0.014943 .

4.2.4 The Fundamental Matrix

The Fundamental Matrix $N = -K^{-1}$ is the centerpiece of the COR model's survival analysis. It translates the instantaneous flow rates of the generator into the expected number of days the system will reside in each state before *leaking* or failing.

$$N = -K^{-1} = \begin{bmatrix} 33.14 & 19.06 & 2.14 \\ 19.66 & 41.77 & 4.70 \\ 12.12 & 25.75 & 36.98 \end{bmatrix}. \quad (5)$$

The diagonal elements n_{ii} represent the total expected time spent in State i (*State Persistence*) given

that the system started in State i . State 2 (Moderate Volatility) has the longest expected residence at 47.88 days, closely followed by State 3 (Crisis) at 45.66 days.

The off-diagonal elements show how much *time* one state contributes to another (*Inter-Regime Support*) before exiting the system. For instance, if the system starts in the Bull regime (State 1), it is expected to spend about 23 days in the Neutral regime (State 2) before eventually *dying* or hitting the ruin threshold. If the system starts in State 3, it only expects to spend 5.63 days in the *safe* State 1, highlighting the difficulty of escaping high-volatility regimes (*Crisis Vulnerability*) during the 2007–2009 period.

Summing the rows of N gives the total expected time until a ruin event occurs (*Total Expected Life*) for each starting state. For a system starting in State 2, the total *life expectancy* is approximately 72.07 days ($17.41 + 47.88 + 6.77$).

4.2.5 The State Summary

The State Summary provides the empirical parameters for each regime (Table 2), bridging the abstract matrix algebra of the COR model with the actual market behavior of the SPX from 2007–2009.

Table 2: State Summary (Student- t HMM)

State	Mean (μ)	Volatility (σ)	Degrees of Freedom (ν)	Expected Spell (Days)
1 (Bull)	1.3559×10^{-3}	0.005699	4	47.01
2 (Neutral)	-2.6188×10^{-4}	0.013952	100	52.09
3 (Crisis)	-2.7099×10^{-3}	0.030853	9	132.14

As tabulated in Table 2, State 1, the *Low Volatility/Bull State* exhibits the highest positive drift and the lowest volatility. Interestingly, it has a low $\nu = 4$, indicating that even in *good* times, the returns had heavy tails. State 2, the *Moderate/Gaussian-like*, with $\nu = 100$, is effectively Gaussian. It represents a *normal* market state with near-zero drift and moderate risk. State 3 (*High Volatility/Crisis*), is characterized by extreme volatility (over $5\times$ higher than State 1) and a significant negative drift. The expected spell of 132.14 days underscores the persistent nature of the 2008 financial collapse. State 3, *Tail Risk* (ϵ and q_{tail}), carries a local tail probability (q_{tail}) of 0.0575, which is nearly 600% higher than the target $\alpha = 0.01$.

The COR model effectively aggregates these distinct states into a single, unified survival outlook for a 250-day trading horizon : the calculated Residence-Weighted Tail Probability ($\bar{q}$) is 0.011956, which represents the average probability of a tail event occurring across the different regimes. This is complemented by a Residence-Weighted Hazard ($\bar{\lambda}$) of 0.014943, reflecting the integrated risk of "leaking" out of a stable state into a ruin event. Ultimately, for a horizon of $T = 250$, the model establishes a Ruin Bound of 0.043681, indicating approximately a 4.37% probability of the system failing to survive the specified threshold over the course of a trading year

4.2.6 Regime Characteristics

The SPX COR analysis reveals a market structure characterized by a three-state regime architecture (Table 3).

Table 3: SPX Student- t regime characteristics.

State	Mean (μ)	Volatility (σ)	Spell Length
1 (Calm)	0.00141	0.00630	51.75
2 (Transitional)	-0.00075	0.01635	43.36
3 (Crisis)	-0.00595	0.04687	54.67

State 1: Crisis (The “Systemic Rupture” Regime)

- **Description:** The Crisis regime is a non-ergodic state where standard valuation models break down.
- **Mathematical Signature:** This state is defined by extreme volatility ($\sigma > 0.046$), significant negative drift ($\mu \ll 0$), and a high concentration of tail risk ($q_3 = 0.14450$).
- **Tail & Hazard Dynamics:** Paradoxically, while this state represents maximum catastrophic intensity, it exhibits the highest degrees of freedom ($\nu = 40$) and the lowest hazard leakage ($\epsilon_3 = 0.0113$) in the SPX system. This indicates that the crisis is *duration-driven*: the system becomes trapped in a high-volatility basin where probability mass remains persistent rather than rapidly leaking out.
- **Behavioral Context:** Identified as a “2008 Liquidity Trap,” this regime is characterized by forced liquidations, repeated limit-down moves, and a collapse of the bid–ask spread.
- **Fragility Status: Maximum.** The system exhibits *Catastrophe Fragility*, where the probability mass is fully concentrated in the “Ruin Zone.”

State 2: Transitional/Steady (The “Escalation” Regime)

- **Description:** The Steady regime is the most deceptive state, appearing stable while underlying instability begins to accumulate.
- **Mathematical Signature:** Characterized by moderate volatility ($\sigma \approx 0.016$) and near-zero or slightly negative drift ($\mu \lesssim 0$).

- **Tail & Hazard Dynamics:** This state functions as a structural conduit with a tail probability ($q_2 = 0.00118$) and moderate degrees of freedom ($\nu = 24$). The hazard leakage ($\epsilon_2 = 0.0121$) exceeds that of the Crisis state, indicating that structural integrity begins to degrade prior to full systemic rupture.
- **Behavioral Context:** Represents the “normalization” phase in which markets transition away from a tranquil bull regime but have not yet entered panic dynamics.
- **Fragility Status: Increasing.** As volatility expands, the probability mass begins to migrate toward the tails, increasing transition intensity ($a_{23} = 0.0033$) toward the Crisis state.

In the COR model (Figure 1), leakage via a_{23} represents the probability mass that exits the modeled system due to systemic rupture, liquidations, or structural failure. For each state i , the leakage is defined by the row sum deficiency: $\epsilon_i = 1 - \sum_{j=1}^3 A_{ij}$. Hence, the value of $a_{23} = 0.0033$ is the critical *gatekeeper* for SPX. While the leakage (ϵ_2) at this stage is almost zero, this transition intensity represents the flow of probability mass from a stable equilibrium into the Crisis state.

State 3: Calm (The “Stationary” Regime)

- **Description:** The Calm regime represents a high-entropy equilibrium state in which the market exhibits stable, mean-reverting dynamics.
- **Mathematical Signature:** Characterized by very low volatility ($\sigma \approx 0.006$) and positive drift ($\mu > 0$), consistent with a locally stable stochastic equilibrium.
- **Tail & Hazard Dynamics:** This regime exhibits tightly concentrated probability mass with negligible tail exposure. Hazard leakage remains minimal ($\epsilon \approx 0$), indicating strong containment within the central region of the distribution and high structural integrity.
- **Behavioral Context:** Represents the “Goldilocks” environment, where liquidity is abundant and price dynamics are dominated by small, approximately Gaussian fluctuations.
- **Fragility Status: Low.** The system resides in a stable basin with minimal probability-mass dispersion, implying low transition pressure toward higher-risk regimes.

4.3 Hazard and Tail Risk Structure

The crisis regime displays significantly elevated tail probability, indicating concentration of extreme downside risk (Table 4). State 1 (Bull) is characterized by the highest positive drift and lowest volatility, though the low degrees of freedom ($\nu = 4$) indicate significant *fat-tailed* risk even during growth periods. State 2 (Neutral) acts as a near-Gaussian state ($\nu = 100$) with moderate volatility

and a slightly negative mean return. State 3 (Crisis) represents the depth of the 2008 financial crisis, exhibiting extreme volatility (0.030853) and the longest expected persistence of 132.14 days.

The COR model effectively aggregates these distinct states into a single, unified survival outlook for a 250-day trading horizon: the calculated Residence-Weighted Tail Probability ($\bar{q}$) is 0.011956, which represents the average probability of a tail event occurring across the different regimes. This is complemented by a Residence-Weighted Hazard ($\bar{\lambda}$) of 0.014943, reflecting the integrated risk of *leaking* out of a stable state into a ruin event. Ultimately, for a horizon of $T = 250$, the model establishes a Ruin Bound of 0.043681, indicating approximately a 4.37% probability of the system failing to survive the specified threshold over the course of a trading year.

4.4 Findings for ES

4.4.1 Model Selection and Identification

This analysis provides a dissection of the Clock of Regimes (COR) analysis for the E-mini S&P 500 Futures (ES) during the 2007–2009 period. This model identifies the survival geometry of market regimes by treating transitions as a *leaky* system. The analysis compares two Hidden Markov Model (HMM) specifications to determine which best captures the statistical properties of the ES returns: The Student- t HMM log-likelihood (2155.14) is significantly higher than the Gaussian HMM (2148.76). This indicates that the ES price action during the financial crisis was characterized by heavy tails and extreme outliers that the Gaussian model fails to accommodate (Table 5).

Table 4: Hazard and tail diagnostics of SPX transition matrix.

State	ν	ϵ_i	Tail Probability q_i
1	4	0.0225	0.00039
2	24	0.0121	0.00118
3	40	0.0113	0.14450

Table 5: ES regime characterization summary under COR framework.

State	Regime	Drift (μ)	Volatility (σ)	Interpretation
1	Bull	1.41×10^{-3}	0.0063	Positive drift with low volatility; represents a stable growth regime with tightly contained probability mass and minimal tail leakage.
2	Neutral	-7.46×10^{-4}	0.0163	Slightly negative drift with moderate volatility; transitional regime where fragility begins to accumulate and probability mass starts to diffuse outward.
3	Crisis	-5.95×10^{-3}	0.0468	Strong negative drift with extreme volatility; crash regime characterized by heavy tails, high hazard intensity, and dominant contribution to tail risk.

4.4.2 Transition Dynamics

Gaussian Transition Matrix A : The Gaussian HMM identifies these transition probabilities:

$$A_{Gaussian} = \begin{bmatrix} 0.9581 & 0.0419 & 0.0000 \\ 0.0102 & 0.9860 & 0.0038 \\ 0.0000 & 0.0140 & 0.9860 \end{bmatrix}. \quad (6)$$

Student- t Transition Matrix A : The Student- t model, which achieved a higher log-likelihood of 2155.14, identifies the following dynamics:

$$A_{Student-t} = \begin{bmatrix} 0.9807 & 0.0193 & 0.0000 \\ 0.0197 & 0.9769 & 0.0033 \\ 0.0000 & 0.0183 & 0.9817 \end{bmatrix}. \quad (7)$$

Both matrices demonstrate strong diagonal dominance ($a_{ii} > 0.95$), indicating that market regimes are highly persistent day-to-day. Both models show a 0.000000 probability for direct transitions between State 1 (Bull) and State 3 (Crisis), reinforcing that the COR model idiom that the system must transition through the *neutral* State 2. In the Student- t model, State 1 is more stable (0.9807) than in the Gaussian model (0.9581), which directly influences the exit rates calculated in the generator K .

4.4.3 Hazard-Adjusted Survival Operator

The Hazard-adjusted survival operator Q for the E-mini S&P 500 Futures (ES) represents the transition dynamics modified by the local survival probabilities of each regime. Based on the provided COR analysis report, the matrix is defined as follows,

$$Q = \begin{bmatrix} 0.9586 & 0.0189 & 0.0000 \\ 0.0195 & 0.9651 & 0.0033 \\ 0.0000 & 0.0181 & 0.9707 \end{bmatrix}. \quad (8)$$

Each row sum satisfies

$$\sum_j q_{ij} = 1 - \epsilon_i.$$

Similar to the SPX model, there is a 0.000000 probability of a direct jump between State 1 (Bull) and State 3 (Crisis), indicating the system must transition through State 2 (*Adjacency Constraints*). The rows sum to less than 1.0 (*Sub-stochastic Property*), where the *missing* probability mass represents the daily hazard or risk of entering a ruin state from that specific regime. State 3 (Crisis) shows the highest persistence in this operator at 0.970664, followed by State 2 (0.965133) and State 1 (0.958612).

4.4.4 The COR Generator Matrix

The COR generator K matrix is derived from the hazard-adjusted transition operator Q as $K = Q - I$. This matrix represents the instantaneous flow rates between regimes and the local *leakage* (risk) or hazard rate of each system state.

$$K = Q - I = \begin{bmatrix} -0.0414 & 0.0189 & 0.0000 \\ 0.0195 & -0.0349 & 0.0033 \\ 0.0000 & 0.0181 & -0.0293 \end{bmatrix}. \quad (9)$$

The diagonal elements (*Exit Rates*) represent the total rate of exit from each state. State 1 has the highest exit rate at -0.041388 , indicating it is the least stable regime in this survival geometry. The off-diagonal values (*Flow Dynamics*) indicate the specific rates of transitioning between states. For instance, the rate of moving from the Neutral State 2 to the Bull State 1 is 0.019483. Because the rows do not sum to zero, the *missing* probability mass (*Hazard Content*) represents the local hazard rate for each state. This contributes to the residence-weighted hazard ($\bar{\lambda}$) of 0.015945.

Table 6: Model fit and statistical characteristics comparison of SPX (Index) and ES (Futures) COR model metrics.

Metric	SPX (Index)	ES (Futures)	Difference
Log-Likelihood (Student- t)	2147.54	2155.14	+7.60
Log-Likelihood (Gaussian)	2140.54	2148.76	+8.22
State 3 Sigma (σ)	0.03085	0.04687	+51.9%
Empirical Tail Threshold	-0.05654	-0.05632	+0.00022

4.4.5 The Fundamental Matrix

The Fundamental Matrix $N = -K^{-1}$ is the centerpiece of the COR model's survival analysis. It translates the instantaneous flow rates of the generator K matrix into the expected number of days the system will reside in each state before *leaking* or failing due to a tail event.

$$N = -K^{-1} = \begin{bmatrix} 33.14 & 19.06 & 2.14 \\ 19.66 & 41.77 & 4.70 \\ 12.12 & 25.75 & 36.98 \end{bmatrix} \quad (10)$$

The diagonal elements N_{ii} represent the total expected time spent in State i given the system started there. State 2 (Neutral) has the highest expected residence at 41.77 days, followed by State 3 (Crisis) at 36.98 days.

The off-diagonal elements show how much *time* probability mass in one state contributes to another (*Inter-Regime Support*) before exiting. For instance, if the system starts in the Crisis regime (State 3), it is only expected to spend 12.12 days in the safer Bull regime (State 1) before hitting the ruin threshold, reflecting the gravitational pull for probability mass of high-volatility states.

Summing the rows of N gives the total expected time until a ruin event occurs (*Total Expected Life*) for each starting state. For a system starting in State 2, the total *life expectancy* is approximately 66.13 days ($19.66 + 41.77 + 4.70$).

4.5 Quantitative COR model comparison of SPX and ES.

A quantitative comparison between the SPX (S&P 500 Index) and ES (E-mini S&P 500 Futures) COR analyses reveals that while both datasets identify three distinct regimes during the 2007–2009 crash period, the ES futures exhibit higher volatility and greater tail risk (Table 6). Both analyses found the Student- t HMM to be superior to the Gaussian HMM in terms of log-likelihood and AIC, confirming that asset returns during this period had significant *fat tails*.

Regime Residence and Stability: The Fundamental Matrix (N) highlights the expected number of days the system resides in each state before *leaking* or failing. The ES futures show shorter expected residence in the Neutral state (41.77 days) compared to the SPX (47.89 days). The SPX spent significantly more time in the High-Volatility *Crisis* regime, with a residence-weight (π) of 20.61% , compared to only 12.25% for the ES futures. In the Generator Matrix (K), the exit rate for the Bull state (State 1) is higher for SPX (-0.0433) than for ES (-0.0414), suggesting slightly less stability in the index’s growth regime during this window.

Risk and Ruin Analysis: The primary distinction between the two datasets lies in their integrated risk metrics over a 250-day horizon. The ES futures Tail Probability ($\bar{q}$) exhibits a residence-weighted tail probability of 0.0184, which is 54% higher than the SPX’s 0.0119. The integrated hazard rate ($\bar{\lambda}$) for ES of 0.0159 exceeds that of the SPX at 0.0149. Finally, the probability of crossing the ruin bound’s threshold over the 250-day horizon is substantially higher for the ES futures market (7.08%) than for the SPX cash index (4.37%).

Comparison of State Summaries (Student- t): Table 7 provides a quantitative comparison of the Regime Means (μ) and Expected Spell Durations (in days) between the S&P 500 Index (SPX) and the E-mini S&P 500 Futures (ES) during the 2007–2009 period. *State 1 (Bull)*: Both assets show positive drift, with ES exhibiting a slightly higher mean return (1.41×10^{-3}) and a longer expected stay of 51.75 days compared to the SPX’s 47.01 days. *State 2 (Neutral)*: This regime is characterized by a slight negative drift in both cases. The SPX remains in this state longer (52.09 days) than the ES futures (43.36 days). *State 3 (Crisis)*: The most significant divergence occurs here. The ES futures experienced a much deeper negative drift (-5.95×10^{-3}) than the SPX (-2.71×10^{-3}). However, the SPX shows a far greater persistence in this regime, with an expected spell of 132.14 days compared to just 54.67 days for the ES.

Table 7: Comparison of Regime Means and Expected Spell Durations: SPX vs ES.

State	SPX Mean (μ)	ES Mean (μ)	SPX Spell (Days)	ES Spell (Days)
State 1 (Bull)	1.35×10^{-3}	1.41×10^{-3}	47.01	51.75
State 2 (Neutral)	-2.62×10^{-4}	-7.46×10^{-4}	52.09	43.36
State 3 (Crisis)	-2.71×10^{-3}	-5.95×10^{-3}	132.14	54.67

Severity vs. Duration: While the ES futures suffered more extreme negative returns during the *Crisis* state (-5.95×10^{-3}) compared to the SPX (-2.71×10^{-3}), the SPX (cash index) tended to remain trapped in that regime for more than twice as long, with an expected spell of 132.14 days versus 54.67 days for the ES (Figure 2).

Market Sentiment: The *Neutral* state acts as a primary residence for the SPX, evidenced by an expected spell of 52.09 days, while the ES futures show more frequent movement between the Bull and Crisis extremes, reflected in shorter Neutral spells of 43.36 days.

Risk Divergence: Table 8 presents a quantitative comparison of the volatility (σ) and tail-risk parameters between the SPX (Index) and ES (Futures) datasets based on the Student- t HMM results. The most dramatic difference occurs in State 3, where the ES futures exhibit over 50% higher volatility than the underlying index. This suggests that during the height of the 2007–2009 crisis, the futures market absorbed significantly more variance and selling pressure than the cash index.

The SPX Bull regime (State 1) has a degree-of-freedom parameter $\nu = 4$ (tail heavyness), matching the ES Bull regime. This indicates that both markets possessed similar *fat tails* even during periods of growth. In the Crisis regime, the ES has a localized tail-event probability (q_{tail}) of 0.144499, which is more than 2.5 times higher than the SPX's 0.057500 in the same state. Due to the higher volatility and local tail risk in the futures, the residence-weighted hazard ($\bar{\lambda}$) for ES is 0.015945, compared to 0.014943 for the SPX.

Comparative Fundamental Matrices (N): The following provides a side-by-side comparison of the Fundamental Matrix ($N = -K^{-1}$) for the SPX and ES datasets to quantify the difference in total *life expectancy* and regime support during the 2007–2009 period.

For the S&P 500 Index (SPX):

$$N_{SPX} = \begin{bmatrix} 31.46 & 23.00 & 3.26 \\ 17.41 & 47.89 & 6.78 \\ 5.64 & 15.51 & 45.66 \end{bmatrix}$$

For the E-mini S&P 500 Futures (ES)

$$N_{ES} = \begin{bmatrix} 33.14 & 19.06 & 2.14 \\ 19.66 & 41.77 & 4.70 \\ 12.12 & 25.75 & 36.98 \end{bmatrix}$$

Summing the rows of N reveals the total expected days until a ruin event occurs. A system

Table 8: Volatility and risk profile comparison: SPX vs ES.

State	SPX Volatility (σ)	ES Volatility (σ)	Relative Difference
1 (Bull)	0.005699	0.006304	+10.6%
2 (Neutral)	0.013952	0.016350	+17.2%
3 (Crisis)	0.030853	0.046875	+51.9%

starting in the Neutral state (State 2) has a life expectancy of 72.08 days for the SPX compared to 66.13 days for the ES, indicating the futures market faced a 9% faster *decay* toward ruin. The diagonal elements show that the SPX remained in a Neutral state (State 2) for an average of 47.89 days, while the ES only managed 41.77 days. If the system starts in the Bull state (State 1), it is expected to spend 3.26 days in the Crisis state (State 3) for SPX, but only 2.14 days for ES. This suggests that while ES regimes are more volatile, the index (SPX) had a stronger *gravitational pull* into prolonged crisis residency. Starting from the Crisis state (State 3), the SPX is expected to spend only 5.64 days in the Bull state before ruin, whereas the ES expects 12.12 days. The futures market showed more *bounce* or mean-reversion potential, even if the moves themselves were more severe.

Ruin Bounds: Table 9 presents a comparative analysis of the Ruin Bounds over a 250-day horizon ($T = 250$) highlighting the increased risk profile of the futures market relative to the cash index during the 2007–2009 period. The ES futures exhibit a significantly higher probability mass in the tails ($\bar{q} = 0.0184$) compared to the SPX index ($\bar{q} = 0.0119$). While the hazard rates ($\bar{\lambda}$) are relatively close, the non-linear expansion of risk over the 250-day horizon results in a 7.09% ruin probability for the ES futures, nearly 1.6 times higher than the SPX’s 4.37%.

Despite both models using a similar empirical left-tail threshold (approximately -0.056), the higher volatility and frequency of extreme shifts in the ES data drive a much more aggressive ruin bound. Because the generator K matrix is an open/leaky operator, these ruin bounds are derived from the residence-weight vector implied by the Fundamental Matrix (N), providing a more realistic long-term risk assessment than static HMM averages.

5 Overview

In the COR model, risk is no longer treated as a one-period variance statistic, but as a temporal survival problem. The central conceptual contribution of the COR model is to reinterpret regime switching as an open system in which probability mass does not merely move across latent states but can also leak out through regime-specific hazard channels. That change is decisive. In ordinary hidden Markov work ????, the focus is usually on filtered state classification, transition persistence, and perhaps volatility segmentation. In the COR framework, however, the object of interest becomes

Table 9: Quantitative comparison of ruin and tail risk: SPX vs ES.

Metric	SPX (Index)	ES (Futures)	% Difference
Residence-Weighted Tail Probability ($\bar{q}$)	0.011956	0.018435	+54.2%
Residence-Weighted Hazard ($\bar{\lambda}$)	0.014943	0.015945	+6.7%
Ruin Bound ($T = 250$)	0.043681	0.070855	+62.2%

the survival-weighted geometry of the system: how long the probability mass remains concentrated in adverse states, how easily it escapes, and how the tail profile of each regime transforms local stress into cumulative ruin exposure. This operator-based shift from static probabilities to temporal exposure is what gives the COR framework its originality and explanatory power.

5.1 Theoretical Architecture and Operator Bridge

The COR model fundamentally reinterprets financial risk as a *leaky* dynamical system where systemic probability mass—defined as the content of probability accumulated over time—dissipates through regime-specific hazard rates. The model transitions through a hierarchical operator structure which is the backbone of the model: $A \rightarrow Q \rightarrow K \rightarrow N$ (Figure ??).

The transition A matrix describes the discrete-time closed-system dynamics of the market. The open transition operator matrix (Q) adjusts the A matrix for *leakage* parameters (ϵ_i), reflecting the probability of systemic structural exit, such as liquidation or market shutdown. The COR generator K matrix represents the continuous-time instantaneous rates of flow and hazard between regimes. The Fundamental Matrix, N , is derived as $N = -K^{-1}$. This matrix calculates the expected residence or duration time (temporal exposure) the market spends in each regime before exiting the system.

The COR architecture allows for a Spectral Fragility Principle, where an increase in the spectral radius of the fundamental matrix, $\rho(N)$, indicates an expansion of survival-weighted residence time in adverse regimes, signaling growing systemic instability.

5.2 Empirical Findings: SPX vs. ES (2007–2009)

Empirical implementation using daily data from the Global Financial Crisis crash confirms that a Student- t HMM specification is statistically superior to Gaussian alternatives. The Student- t model achieved significantly higher log-likelihoods for both the S&P 500 Index (2147.54) and E-mini S&P 500 Futures (2155.14), highlighting the necessity of capturing *fat tails* to evaluate catastrophe fragility.

Our study identified three distinct states: Stress/Crisis, Steady/Neutral, and Calm/Bull. Table 10 presents a quantitative comparison revealing divergent failure pathways for these structurally similar markets.

5.3 Fragility Decomposition and Ruin Bounds

The COR framework decomposes financial instability into two primary channels (Figure 2): (i) *Duration Fragility*, which arises when a system remains persistently trapped in unfavorable conditions (e.g., the SPX's prolonged 132-day crisis spell), and (ii) *Catastrophe Fragility*, which reflects extreme

tail losses that produce rapid destruction over short horizons (e.g., the high-intensity -5.95×10^{-3} drift in ES).

The model aggregates these factors into a *Residence-Weighted Ruin Bound*. Despite similar empirical left-tail thresholds (≈ -0.056), the ES futures market demonstrated a more aggressive ruin bound (0.070855) compared to the SPX index (0.043681). This disparity stems from the futures market’s higher residence-weighted tail probability ($\bar{q} = 0.0184$), which is 54% higher than that for the SPX index.

6 Discussion

The COR model provides a coherent and ambitious interpretation of financial fragility in which risk is no longer treated as a one-period variance statistic, but as a temporal survival problem. The Talebian view, as articulated by Nassim Nicholas Taleb [?](#), is a philosophical and mathematical framework centered on how systems handle uncertainty, disorder, and *Black Swan* events [?](#). It rejects conventional Gaussian risk models in favor of a structural approach to fragility and ruin. The COR model closely aligns with the Talebian view by shifting the focus from thin-tailed, linear

Table 10: Comparative Regime Dynamics: SPX vs ES.

Metric	SPX (Index)	ES (Futures)	Interpretation
Crisis Drift (μ)	-2.71×10^{-3}	-5.95×10^{-3}	ES suffered more extreme negative returns during the crisis regime.
Crisis Persistence	132.14 days	54.67 days	SPX was trapped in the crisis regime for more than twice as long, indicating stronger duration fragility.
Crisis Volatility (σ)	0.03085	0.04687	ES futures exhibited 51.9% higher crisis-state volatility, implying greater catastrophe intensity.
Ruin Probability ($T = 250$)	4.37%	7.09%	ES faced approximately 1.6 times the probability of systemic failure over the 250-day horizon.

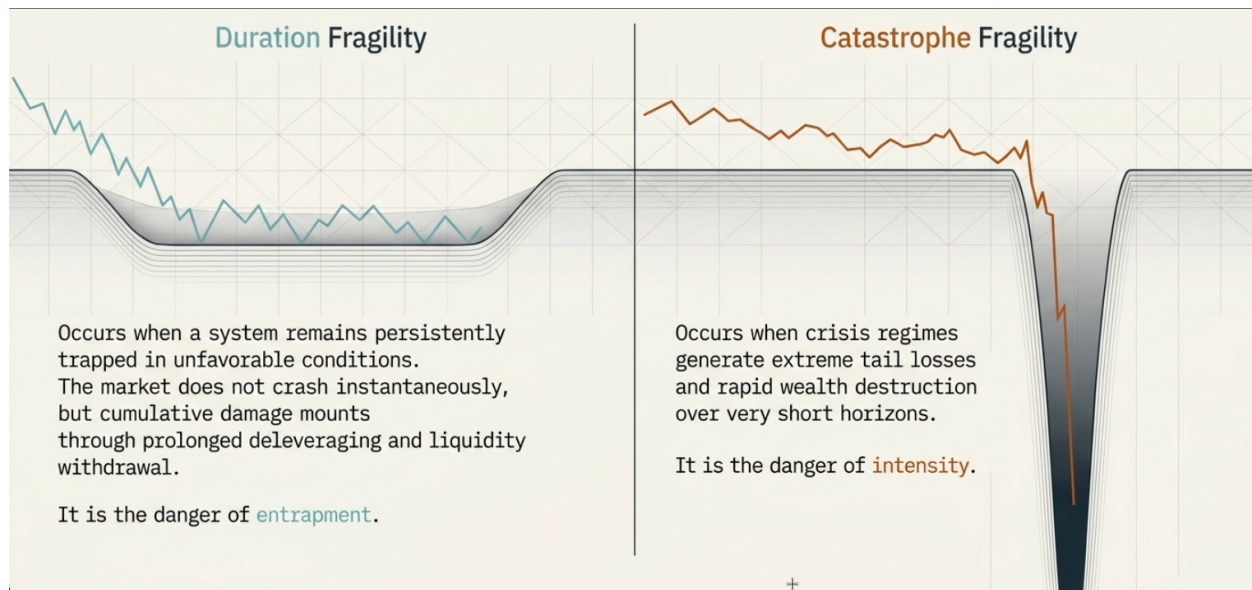


Figure 2: Duration Fragility versus Catastrophe Fragility.

risk metrics to the nonlinear, temporal reality of systemic fragility. In this framework, risk is not a static *one-period variance statistic* like Standard Deviation or Value-at-Risk (VaR), but a survival-weighted accumulation of exposure to ruin.

We framed fragility through two distinct channels, and that decomposition is one of the strongest parts of our approach. *Duration fragility* refers to prolonged residence in adverse regimes (Figure 2). In practical terms, this is the case where a market does not necessarily crash instantaneously, but becomes trapped in a stressed condition long enough for cumulative damage to mount through deleveraging, liquidity withdrawal, credit contraction, or repeated losses. *Catastrophe fragility*, by contrast, refers to states whose tail distributions are so severe that large losses can materialize very quickly, even without long persistence (Figure 2). The importance of the COR model is that it can represent both mechanisms simultaneously. A system may be fragile because it remains too long in a moderately bad regime, or because it briefly touches a highly destructive tail regime, or because both effects reinforce one another. Standard risk models often blend these channels together. The COR model cleanly separates them. The COR model demonstrated that while futures markets (ES) absorb more extreme *catastrophic* shocks, cash indices (SPX) may exhibit greater *duration* fragility through prolonged entrapment in high-volatility basins.

Our paper highlights two major shortcomings in financial econometrics. The first issue is the reliance on Gaussian or diffusion-style assumptions, which often downplay extreme observations and treat volatility as too stable. The second concern revolves around the use of standard Hidden Markov Models (HMMs) with geometric residence-time structures, which can underestimate the clustering and persistence of crisis conditions. This critique is valid. In crisis data, it's important

to recognize that not only are returns fat-tailed, but time itself also becomes asymmetric. Stress episodes cannot be treated as interchangeable with calm episodes simply because they occupy the same time unit. A week during a crisis has a different economic significance compared to a week in a calm period. The COR model formalizes this concept by defining probability mass as temporally accumulated probability, rather than just an instantaneous measure.

The open transition operator Q is especially important because it formalizes the notion that regime switching alone is not the whole story; the system is also subject to survival loss. In economic terms, the hazard vector ϵ acts like a structural permeability parameter. Higher hazard means the regime is less capable of containing probability mass and more likely to transmit it into systemic failure, absorption, or external ruin. This is why the rows of Q sum to less than one. The missing probability mass is not numerical error. It is the model's explicit representation of openness.

The generator identification result is also conceptually useful because it places the model within a broader continuous-time tradition. Even if the empirical estimation begins with a discrete HMM, the transition to K and N gives access to tools familiar from compartmental analysis ?????, survival theory ??, and continuous-time Markov processes ??????. That is a strong interdisciplinary move. It allows our work to speak not only to financial econometrics but also to mathematical systems theory. The operator bridge theorem makes that explicit: the model progressively translates persistence into hazard-adjusted survival, then into generator form, and finally into residence-time (duration) structure. Each step changes the interpretation of the same underlying market process without discarding information.

The fragility decomposition theorem is one of the clearest summaries of the model's intuition, as demonstrated in Theorem A.4 (Fragility Decomposition) in the Mathematical Appendix. Fragility is treated as the interaction of residence mass, tail severity, and hazard leakage. This multiplicative logic is economically sensible. Residence mass captures temporal exposure. Tail severity captures the intensity of adverse outcomes conditional on being in a state. Hazard leakage captures the openness of the system and thus its vulnerability to structural loss. None of these alone is sufficient. A regime can have heavy tails but low residence mass, or long residence but mild tails, or high hazard but little persistence. True systemic fragility emerges when all three are sufficiently active together. That formulation is one of our work's most useful contributions because it explains why similar markets can fail differently.

The spectral fragility principle extends the framework beyond simple summary metrics as demonstrated in Theorem A.5 (Spectral Radius and Norm Bound) in the Mathematical Appendix. By linking fragility to the spectral radius of N , we propose an operator-level early-warning concept. A larger $\rho(N)$ means an expansion of survival-weighted residence time, which in turn means that adverse structures are becoming more persistent or more difficult to escape. This is appealing because it makes fragility a property of the whole system, not just a property of one regime parameter.

Spectral diagnostics are particularly attractive in crisis analysis because they may move before realized collapse occurs. In that sense, the COR model is not merely descriptive; it has potential as a monitoring device.

Using Student- t emissions for the latent regimes is essential. Heavy-tailed distributions are not optional in crisis-era data, and the use of regime-specific degrees of freedom ν_i allows the tail profile itself to vary across market states. This is important because calm, transitional, and crisis conditions do not differ only in mean and volatility; they may differ in the shape of their tails. The hazard calibration $\epsilon_i = \epsilon_{\min} + c/\nu_i$ is elegant because it ties system openness directly to tail thickness. Regimes with smaller ν_i are more heavy-tailed and therefore more structurally leaky. That mapping gives the model a tight internal logic: tail behavior influences hazard, hazard influences Q , Q influences K , and K determines the temporal geometry through N .

The results for SPX strongly support the paper's theoretical claims. First, the Student- t HMM outperforms the Gaussian model in log-likelihood and AIC, which confirms that heavy tails matter materially for the 2007–2009 period. This is a crucial empirical validation of the model's premise. The gain in fit is not merely statistical housekeeping; it indicates that crisis returns cannot be adequately described by thin-tailed within-regime shocks. The paper rightly interprets this as evidence that *catastrophe fragility* is an intrinsic feature of the data.

Second, the transition matrices reveal a highly persistent three-state regime structure. All diagonal elements exceed 0.95, so the system is sticky in every state. That alone would already be important, but the absence of direct transitions between the calm and crisis states is even more informative. It means that the market generally moves through an intermediate regime rather than jumping directly from tranquility to rupture. This supports the interpretation of the middle state as a conduit of fragility accumulation. In COR model terms, the system is path-dependent: deterioration tends to occur through an escalation channel rather than through instantaneous discontinuity.

The hazard-adjusted operator Q deepens that interpretation. Because its rows sum to less than one, the market is explicitly modeled as a leaky survival system. The missing row mass is the local hazard content, which makes the state process economically richer than a standard Markov chain. The generator K then shows which states are more transient and which flows are more relevant. In the SPX case, the bull state has the highest exit rate, suggesting that calm growth is relatively easy to leave during the crisis sample. That is a telling result: the safe regime is structurally fragile, even if its one-step volatility is low.

6.1 S&P 500 Index (SPX)

The SPX fundamental matrix is especially revealing. Its diagonal entries indicate substantial residence in all three states, but particularly in the transitional and crisis states. The interpretation is powerful: the market during 2007–2009 was not merely volatile; it was trapped. This is the

empirical manifestation of duration fragility. The system did not only experience bad states; it accumulated time within them. Off-diagonal entries add further nuance by showing how one state supports residence in another before exit. The neutral state, in particular, appears to function as the main staging area through which adverse persistence is transmitted.

The SPX state summary illustrates the economic meaning of the latent regimes. The bull state combines positive drift with low volatility, but still has heavy tails. This is important because it shows that apparently favorable conditions are not free of fragility. The neutral state is close to Gaussian and serves as the dominant transitional basin. The crisis state combines high volatility, negative drift, and long expected spell duration, which makes it the natural locus of duration fragility. The very long expected spell in the crisis state is a critical empirical finding. It says that during the financial crisis, systemic stress in the cash index was not only severe; it was persistent enough to become structurally entrapping.

The SPX hazard and tail diagnostics reinforce this picture. The crisis state has by far the largest tail probability, so *catastrophe risk* is concentrated there. Yet the hazard structure also matters. The interplay between ν , ϵ_i , and q_i shows that the most dangerous state is not simply the one with the worst volatility reading, but the one where heavy tails, persistence, and leakage jointly support ruin accumulation. That is precisely the advantage of the COR model framework: it converts descriptive regime labels into a measurable structural mechanism.

The aggregate SPX measures summarize this mechanism cleanly. The residence - weighted tail probability $\bar{q} = 0.011956$, the residence-weighted hazard $\bar{\lambda} = 0.014943$, and the one-year ruin bound of 0.043681 provide a compact survival-based view of systemic risk. Their interpretation should be handled carefully. These are not plain unconditional probabilities in the usual sense; they are structural summaries induced by open-system survival dynamics. The ruin bound, in particular, is best understood as an upper-bound style measure of systemic vulnerability over the horizon, not as a literal point forecast of collapse. Even so, a value around 4.37% is substantial for a broad market index and confirms that the 2007–2009 period involved a nontrivial structural probability of severe degradation.

6.2 E-mini S&P 500 Futures (ES)

The ES results are equally important, and perhaps even more informative from a comparative perspective. Again the Student- t model beats the Gaussian specification, with an even larger log-likelihood. This confirms that futures returns are even more strongly shaped by heavy tails and outliers. That fits the economic intuition that the ES, because of leverage, round-the-clock trading, and liquidity concentration, should absorb stress more violently than the underlying cash index. The paper correctly interprets this as evidence that the futures market is structurally more exposed to explosive tail dynamics.

The ES regime characterization makes that clear. The bull state is similar to SPX in sign and scale, but the neutral state is more negative and the crisis state is dramatically more severe. The crisis-state volatility is much larger than in SPX, and the negative drift is considerably deeper. This suggests that when the futures market enters the crisis regime, it experiences a sharper catastrophe channel. In other words, ES appears more catastrophe-fragile than SPX. The market does not merely become volatile; it becomes violently nonlinear.

Yet the transition and residence results show that ES is not simply *worse* in every dimension. Its crisis-state spell duration is much shorter than SPX's, and its fundamental matrix indicates less prolonged residence in crisis. This is where the paper's decomposition becomes very useful. ES is more severe but less durationally trapped. SPX is less severe but more durationally trapped. Thus the two markets fail through different fragility pathways. ES expresses *catastrophe fragility* more strongly; SPX expresses *duration fragility* more strongly. That is one of the most valuable empirical conclusions in this paper.

The comparative tables support this reading well. The higher State 3 volatility in ES, the similar empirical tail threshold, and the much larger residence-weighted tail probability all indicate that futures carry much stronger extreme-risk intensity. At the same time, the much longer SPX crisis spell shows that the index absorbs stress through persistence and prolonged malaise. This is an important distinction for practitioners. A leveraged or derivative market may destroy wealth faster, while the cash benchmark may trap capital for longer. Both are fragile, but in different ways.

The comparative fundamental matrices deepen the interpretation. ES has shorter total expected life starting from the neutral state, which suggests faster decay toward ruin. But ES also shows more ability to spend some time in the bull state even after starting in crisis, implying more bounce or snapback potential. That too is economically plausible. Futures markets are often more violent in both directions: they can crash faster but also rebound faster. SPX, by contrast, appears to retain a stronger gravitational pull into prolonged crisis residency. This is a beautiful illustration of how the same broad market ecosystem can contain two distinct temporal architectures of risk.

The comparative ruin metrics bring the whole argument together. ES has a residence-weighted tail probability more than 54% higher than SPX, a slightly higher residence-weighted hazard, and a ruin bound over 62% higher. The modest difference in hazard but large difference in ruin bound is especially interesting. It implies that nonlinear amplification is being driven primarily through the tail channel rather than through much larger openness per se. In other words, the futures market is not vastly more fragile because it leaks much more often, but because when it is stressed, the consequences are far more severe (catastrophe-fragility).

6.3 The COR Model: A New Class of Regime Switching Model

The COR model represents a new class of regime-switching model that reframes financial fragility through an open-system operator lens. We examined how classical Markov killing or defective Markov chain models relate to our COR model. In the classical baseline HMM ??, persistence is modeled in discrete time via the transition probability a_{ii} . The system is closed and conservative—trapped inside a stochastic matrix A where rows must sum to exactly 1.0 ($\sum_j a_{ij} = 1$). There is no mechanism for the system itself to terminate, decay, or “die.” The only way out of a state is to transition into another surviving state.

On the other hand, a Markov chain with killing is precisely the mathematical precursor that justifies our COR operator framework. In a Markov killing model, a standard Markov process operates normally until an unobservable, exogenous random time ζ (the “killing time”) occurs, at which point the process is instantly annihilated or sent to a “graveyard state” Δ ?. Mathematically, this transforms a conservative generator into a sub-stochastic or open generator matrix K , where the rows sum to less than zero (in continuous time):

$$\sum_{j \in \mathcal{S}} K_{ij} = -q_i \leq 0,$$

where q_i is the killing rate (or hazard intensity) of state i . If the system is in state i , it has a continuous risk of being killed out of existence rather than merely transitioning.

Why then is the COR model a distinct, new class of regime switching model? While the COR model inherits the underlying probability space of a continuous-time Markov chain with killing, it moves beyond classical survival analysis by establishing the Fundamental Matrix $N = -K^{-1}$ as the primary analytical operator for financial fragility. We can explicitly draw the distinctions that highlight the COR model’s novelty by examining Table 11.

Table 11: Conceptual comparison between Classical HMMs, Markov Killing Models, and the COR Framework

Feature	Classical HMM (Hamilton)	Markov Killing Models	COR Model Framework (KRONX)
System Boundary	Closed (Conservative)	Open (Annihilated)	Open with Observable Leakage
Core Operator	Transition Matrix A	Sub-stochastic Generator K	Fundamental Residence Operator $N = -K^{-1}$
View of Time	Implicit step-counts	Abstract killing time ζ	Continuous-time residence mass allocated across regimes
Risk Application	Volatility clustering	Credit default / structural ruin	Heavy-tailed tail-risk leakage (q) mapped to structural duration (N)

Therefore, as tabulated in Table 11, a claim that the COR model is not new is not supported by the facts. While classical Markov killing models introduce an open boundary via a defective sub-

generator matrix K to model state “killing/annihilation,” the COR framework shifts the analytical focus entirely to the fundamental operator $N = -K^{-1}$. The COR model takes the fundamental matrix N out of traditional absorbing Markov chain theory and turns it into an active tool to measure heavy-tailed financial fragility over a risk horizon. By doing so, it maps the continuous-time residence probability mass of the system directly to heavy-tailed absorption channels, rendering the exact path dependency of structural ruin observable and computationally tractable.

6.4 The Great Financial Crisis — What is New?

Our new contribution regarding the Great Financial Crisis is that it reframes 2007–2009 not merely as a volatility crisis, liquidity crisis, or regime-switching episode, but as a survival-geometry problem: probability mass became trapped, leaked, and accumulated ruin exposure through time.

The main mathematical novelty of this framework lies in the explicit COR decomposition of the Global Financial Crisis (GFC) systemic fragility into two distinct, observable failure channels: *duration fragility* and *catastrophe fragility*. Under the duration channel, the empirical evidence demonstrates that the S&P 500 Index (SPX) did not simply experience a localized, transient asset crash; rather, its underlying Markov chain became deeply trapped within a persistent crisis basin. The framework’s key empirical insight reveals that SPX exhibited a prolonged expected crisis spell duration of approximately 132.14 days, highlighting an unexpected structural entrapment that traditional short-memory risk models fail to capture. Conversely, the catastrophe channel unmasks the severe tail intensity hidden within E-mini S&P 500 (ES) futures. Although ES spent fewer expected days trapped in the crisis state, it manifested substantially sharper localized volatility—exceeding that of SPX by approximately 51.9%—which translated directly into a significantly higher probability mass leakage. This structural vulnerability is further reflected in the model’s ultimate ruin metrics, where ES reached a devastating ruin bound of 7.09% compared to the more contained 4.37% observed for SPX over a 250-day risk horizon.

Hence, our main GFC finding is that SPX failed by duration; ES failed by catastrophe. That is new because standard GFC analyses usually emphasize leverage, mortgage credit, liquidity spirals, volatility, or systemic contagion. This paper adds an operator-theoretic distinction: the cash index absorbed crisis through prolonged residence, while the futures market absorbed crisis through sharper tail-risk leakage.

The second novelty is methodological: our work extends the HMM workflow,

$$A \rightarrow Q \rightarrow K \rightarrow N,$$

where A is the standard transition matrix, Q introduces leakage, $K = Q - I$ creates the open-system generator, and $N = -K^{-1}$ measures expected residence time before ruin. This turns the GFC into

a measurable survival system rather than a closed regime-switching process.

The third novelty is the ruin bound. Instead of stopping at regime probabilities or expected durations, our approach computes residence-weighted tail exposure, residence-weighted hazard, and a finite-horizon ruin probability. This makes the GFC interpretable as a cumulative survival failure, not merely a sequence of large fluctuations of returns.

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The Clock of Regimes Project (CORP) is an advanced statistical finance workflow project whose specific aim is systemic market risk identification and classification, regime-switching tracking, and financial fragility analysis. The project bridges continuous-time matrix operator mathematics with machine learning to identify structural macro shifts and capture non-Gaussian “fat-tailed” tail risks.

Mathematical Appendix

A Open Transition Operator

Theorem A.1 (Hazard-Adjusted Transition Operator). *Let A be a $K \times K$ stochastic transition matrix representing a closed regime system, where $\sum_{j=1}^K A_{ij} = 1$ for all i . Define the hazard-adjusted transition operator Q as:*

$$Q = \text{diag}(1 - \epsilon_1, \dots, 1 - \epsilon_K)A \quad (11)$$

where $\vec{\epsilon} = (\epsilon_1, \dots, \epsilon_K)$ is a regime-dependent hazard vector with $0 \leq \epsilon_i \leq 1$ for each i .

Then, Q is a substochastic matrix such that each row sum satisfies:

$$\sum_{j=1}^K Q_{ij} = 1 - \epsilon_i \leq 1 \quad (12)$$

This construction transforms the regime-switching process into an open Markov system, where the evolution of the system tracks survival-weighted probability mass and accounts for potential absorption outside the modeled regime system.

Proof. The proof that the hazard-adjusted transition operator Q is a substochastic matrix proceeds as follows:

1. **Properties of the Stochastic Matrix A :** By definition, the transition matrix A is a stochastic matrix representing a closed regime system. All entries A_{ij} are non-negative, and each row i sums to unity:

$$\sum_{j=1}^K A_{ij} = 1, \quad \forall i \in \{1, \dots, K\} \quad (13)$$

2. **Definition of Q :** The operator Q is defined by scaling the rows of A by a regime-dependent hazard vector $\vec{\epsilon}$:

$$Q = \text{diag}(1 - \epsilon_1, \dots, 1 - \epsilon_K)A \quad (14)$$

where each leakage parameter satisfies $0 \leq \epsilon_i \leq 1$. Consequently, each entry $Q_{ij} = (1 - \epsilon_i)A_{ij}$.

3. **Row Summation of Q :** Consider the sum of the entries in the i -th row of Q :

$$\sum_{j=1}^K Q_{ij} = \sum_{j=1}^K (1 - \epsilon_i)A_{ij} = (1 - \epsilon_i) \sum_{j=1}^K A_{ij} \quad (15)$$

Substituting the stochastic property $\sum_j A_{ij} = 1$, we obtain:

$$\sum_{j=1}^K Q_{ij} = (1 - \epsilon_i) \times 1 = 1 - \epsilon_i \quad (16)$$

4. **Substochastic Property:** Given the constraint $0 \leq \epsilon_i \leq 1$, it follows that $0 \leq 1 - \epsilon_i \leq 1$. Therefore, the row sum of Q is bounded:

$$\sum_{j=1}^K Q_{ij} = 1 - \epsilon_i \leq 1 \quad (17)$$

5. **Conclusion:** The matrix Q is substochastic because all its entries are non-negative and the sum of each row is less than or equal to 1. This reduction by ϵ_i represents probability mass leaving the system, reflecting the possibility of absorption outside the modeled regime system. Thus, Q describes an open Markov system tracking survival-weighted probability mass.

□

Theorem A.2 (Generator Proxy Identification). *Let Q be the hazard-adjusted transition operator of*

a discrete-time regime system, and let I be the identity matrix. Define the generator matrix K as:

$$K = Q - I \quad (18)$$

For a small time step Δt , the matrix K serves as a discrete-time approximation of the continuous-time Markov generator. This construction establishes an operator bridge linking discrete Hidden Markov Models (HMMs) to continuous-time survival processes via the fundamental relation $N = -K^{-1}$.

Proof. The role of the generator matrix K as a bridge between discrete Hidden Markov Models and continuous-time survival processes is established as follows:

1. **Continuous-Time Dynamics:** In a continuous-time Markov process, the transition probability matrix over an interval t is defined by the matrix exponential:

$$P(t) = e^{Qt} \quad (19)$$

For sufficiently small t , the first-order Taylor expansion provides the linear approximation:

$$e^{Qt} \approx I + Qt \quad (20)$$

2. **Discrete-Time Approximation:** For a discrete-time step Δt , the transition matrix P of a Hidden Markov Model can be approximated using the rate matrix Q :

$$P \approx I + Q\Delta t \quad (21)$$

Substituting the definition of the COR generator $K = Q - I$ (or $Q = K + I$) into the approximation yields:

$$P \approx I + (K + I)\Delta t = I + K\Delta t + I\Delta t \quad (22)$$

This confirms that the structure of K dictates the infinitesimal flow of the system[cite: 1, 2].

3. **Connection to HMMs:** The generator K approximates the behavior of a continuous-time process within a discrete framework. This allows Hidden Markov Models to represent underlying continuous-time dynamics, facilitating the analysis of market processes that transition over potentially infinitesimal time intervals.
4. **Conclusion:** The matrix K , derived from the hazard-adjusted operator Q , acts as a functional bridge between discrete and continuous-time models. It provides a rigorous pathway to

analyze continuous-time survival processes—such as regime-specific residence times and ruin probabilities—through the lens of discrete-time HMMs.

This completes the proof that K serves as the infinitesimal generator proxy for the open regime system. $\square$

Theorem A.3 (Fundamental Matrix and Survival Geometry). *The fundamental matrix of the open system is defined by the operator:*

$$N = -K^{-1} = \int_0^{\infty} e^{Kt} dt \quad (23)$$

The element N_{ij} represents the expected survival-weighted time spent in regime j when starting from regime i . Thus, N provides a direct measure of temporal probability content.

In financial terms, large diagonal entries of N indicate durational entrapment in adverse regimes, while large off-diagonal entries reveal conduit states through which systemic fragility propagates. The fundamental matrix $N = -K^{-1}$ may be derived by inverting the negative of the K matrix. The inverse of this matrix calculates the expected persistence of probability mass (residence time) within each node—essentially, the total time the market spends in a specific regime before it eventually leaks out or transitions.

Proof. To establish the equivalence $N = -K^{-1} = \int_0^{\infty} e^{Kt} dt$, we proceed through the properties of the matrix exponential and its convergence requirements:

1. **Matrix Exponential Definition:** The matrix exponential e^{Kt} is defined via the power series expansion, analogous to the scalar case:

$$e^{Kt} = I + Kt + \frac{(Kt)^2}{2!} + \frac{(Kt)^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{(Kt)^n}{n!} \quad (24)$$

where I denotes the $K \times K$ identity matrix.

2. **Integration of the Matrix Exponential:** Consider the improper integral of the matrix exponential from 0 to ∞ :

$$\int_0^{\infty} e^{Kt} dt = \int_0^{\infty} \left(I + Kt + \frac{(Kt)^2}{2!} + \dots \right) dt \quad (25)$$

3. **Term-by-Term Integration and Convergence:** Integrating the series term-by-term, we observe:

$$\int_0^{\infty} I dt = [tI]_0^{\infty}, \quad \int_0^{\infty} Kt dt = \left[\frac{t^2}{2} K \right]_0^{\infty} \quad (26)$$

These terms appear to diverge unless the matrix K is negative definite or stabilizing (i.e., all eigenvalues have negative real parts). In the COR model, $K = Q - I$ is an open/leaky generator proxy, ensuring that the eigenvalues of K are strictly negative, which guarantees the decay of e^{Kt} as $t \rightarrow \infty$.

4. **Convergence and Inversion:** Under the condition that K is invertible and its eigenvalues have negative real parts, we use the fundamental theorem of calculus for matrices:

$$\frac{d}{dt}e^{Kt} = Ke^{Kt} = e^{Kt}K \quad (27)$$

Integrating both sides from 0 to ∞ :

$$\int_0^\infty Ke^{Kt} dt = [e^{Kt}]_0^\infty \quad (28)$$

Since $\lim_{t \rightarrow \infty} e^{Kt} = 0$ (due to negative eigenvalues) and $e^{K(0)} = I$:

$$K \int_0^\infty e^{Kt} dt = 0 - I = -I \quad (29)$$

5. **Conclusion:** Premultiplying by K^{-1} , we obtain:

$$N = \int_0^\infty e^{Kt} dt = -K^{-1} \quad (30)$$

The result $N = -K^{-1}$ is well-defined and holds provided K is transient, reflecting the expected persistence of probability mass within the regime system before structural exit or ruin.

□

Theorem A.4 (Fragility Decomposition). *In a financial market characterized by K regimes, let q_i denote the left-tail probability of returns conditional on regime i , ϵ_i denote the hazard leakage for regime i , and π_i be the survival-weighted probability of regime i such that $\sum_{i=1}^K \pi_i = 1$. The systemic risk measures are decomposed as follows:*

1. **Residence-weighted tail exposure ($\bar{q}$):**

$$\bar{q} = \sum_{i=1}^K \pi_i q_i \quad (31)$$

2. Average hazard leakage ($\bar{\lambda}$):

$$\bar{\lambda} = \sum_{i=1}^K \pi_i \epsilon_i \quad (32)$$

This decomposition illustrates that financial markets exhibit fragility through two primary channels: duration fragility, where the system remains for extended periods in adverse regimes (π_i), and catastrophe fragility, where losses mount rapidly during crisis episodes (q_i). These weighted sums provide a structural measure of how individual regimes contribute to systemic risk, linking heavy-tailed shocks and regime persistence to the operator-level survival geometry.

Proof. The derivation of the residence-weighted tail exposure ($\bar{q}$) and the average hazard leakage ($\bar{\lambda}$) as systemic risk measures relies on the properties of weighted expectations over the survival-weighted regime distribution π .

1. **Derivation of Residence-Weighted Tail Exposure ($\bar{q}$):** By definition, the systemic tail exposure is the expectation of the regime-specific tail probabilities q_i over the long-run allocation of probability mass. Given that π is the survival-weighted distribution derived from the fundamental matrix N , such that:

$$\sum_{i=1}^K \pi_i = 1, \quad \pi_i \geq 0 \quad (33)$$

The aggregate measure $\bar{q}$ is constructed as the convex combination:

$$\bar{q} = \sum_{i=1}^K \pi_i q_i \quad (34)$$

This weighted average represents the expected probability of a tail event occurring at any given time, accounting for the varying durations the system spends in each latent state.

2. **Derivation of Average Hazard Leakage ($\bar{\lambda}$):** Similarly, let ϵ_i represent the instantaneous hazard leakage (or "openness") of regime i . The systemic hazard rate $\bar{\lambda}$ is the ensemble average of these local leakages across the survival geometry:

$$\bar{\lambda} = \sum_{i=1}^K \pi_i \epsilon_i \quad (35)$$

Since π reflects the expected residence time in each node, $\bar{\lambda}$ quantifies the integrated rate at which probability mass exits the modeled system (ruin intensity).

Conclusion: Both $\bar{q}$ and $\bar{\lambda}$ are valid structural descriptors of the open Markov system. They facilitate the mapping of discrete regime dynamics into a unified survival-oriented risk bound, $P(\text{ruin}) \leq 1 - e^{-T_h \Lambda}$, where Λ is a function of these residence-weighted averages. $\square$

Theorem A.5 (Spectral Radius and Norm Bound). *Let $A \in \mathbb{C}^{n \times n}$ be a square matrix, and let $\rho(A)$ denote its spectral radius, defined as the maximum absolute value of its eigenvalues:*

$$\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\} \quad (36)$$

For any consistent matrix norm $\|\cdot\|$, the spectral radius is bounded by the norm of the matrix:

$$\rho(A) \leq \|A\| \quad (37)$$

This property ensures that the spectral radius provides a lower bound for all matrix norms and is a critical measure for analyzing the stability, convergence, and structural behavior of linear operators in open dynamical systems.

Proof. To prove that the spectral radius $\rho(A)$ of a square matrix $A \in \mathbb{C}^{n \times n}$ is bounded by any consistent matrix norm $\|A\|$, we proceed as follows:

1. **Eigenvalue and Eigenvector Relationship:** Let λ be any eigenvalue of the matrix A , and let v be its corresponding non-zero eigenvector ($v \neq 0$). By the definition of the eigenvalue problem, we have:

$$Av = \lambda v \quad (38)$$

2. **Applying the Matrix Norm:** Taking the norm of both sides of the equation and utilizing the property of absolute homogeneity ($\|\alpha v\| = |\alpha| \cdot \|v\|$ for any scalar α), we obtain:

$$\|Av\| = \|\lambda v\| = |\lambda| \cdot \|v\| \quad (39)$$

3. **Consistency of the Matrix Norm:** By the definition of a consistent matrix norm (submultiplicativity with respect to a vector norm), the following inequality holds for any vector v :

$$\|Av\| \leq \|A\| \cdot \|v\| \quad (40)$$

4. **Combining the Inequalities:** Substituting the expression from Step 2 into the inequality from Step 3, we have:

$$|\lambda| \cdot \|v\| \leq \|A\| \cdot \|v\| \quad (41)$$

Since v is an eigenvector, $\|v\| \neq 0$. Dividing both sides by $\|v\|$ yields:

$$|\lambda| \leq \|A\| \quad (42)$$

5. Conclusion for the Spectral Radius: Since the inequality $|\lambda| \leq \|A\|$ holds for every eigenvalue $\lambda \in \sigma(A)$, it must also hold for the maximum absolute value of those eigenvalues. Therefore:

$$\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\} \leq \|A\| \quad (43)$$

This completes the proof that the spectral radius of a matrix is bounded above by any consistent matrix norm. $\square$

Theorem A.6 (Residence-Weighted Ruin Probability Bound). *Let T_h denote the risk horizon (e.g., $T_h = 250$ days), and let q be the vector of regime-specific tail-loss probabilities. Given the survival-weighted regime distribution π and the fundamental matrix $N = -K^{-1}$, the structural ruin probability for the open dynamical system is bounded by:*

$$P(\text{ruin}) \leq 1 - \exp\left(-T_h \frac{\pi^\top N q}{\pi^\top N \mathbf{1}}\right) \quad (44)$$

where $\mathbf{1}$ is the $K \times 1$ summation vector. This bound captures the non-linear interaction between regime duration, tail exposure, and the systemic hazard structure.

Proof. To provide a rigorous derivation of the residence-weighted ruin probability bound, we integrate elements of stochastic processes, survival analysis, and operator theory. The proof is structured as follows:

1. **Definition of Parameters:** Let T_h denote the risk horizon. We define the following vectors and operators:
 - π : The survival-weighted regime distribution (weights).
 - N : The fundamental matrix ($N = -K^{-1}$) capturing expected residence times.
 - q : The hazard vector of regime-specific tail-exposure rates.
 - $\mathbf{1}$: A $K \times 1$ summation vector of ones.
2. **System Dynamics and Survival:** We model the regime switching as an open Markov process. Let $S(t)$ be the survival function, representing the probability that the system has not encountered a ruin event by time t . The instantaneous transitions and “leakage” are governed by the generator K , where N represents the integrated path of the system until exit.

3. **Derivation of the Cumulative Hazard:** The cumulative hazard over the horizon T_h is determined by the temporal content of probability mass residing in adverse states. In the COR framework, this is expressed as the product of the residence distribution and the tail-exposure rates:

$$H(T_h) = \int_0^{T_h} \pi^\top N q dt \quad (45)$$

Under the assumption of stationary regime dynamics within the horizon, the survival probability follows an exponential decay:

$$S(T_h) = \exp \left(- \int_0^{T_h} \pi^\top N q dt \right) \quad (46)$$

4. **Establishing the Normalized Bound:** To quantify the effective ruin intensity, we normalize the catastrophe volume ($\pi^\top N q$) by the total expected system life expectancy ($\pi^\top N \mathbf{1}$). This ratio defines the residence-weighted hazard rate Λ :

$$\Lambda = \frac{\pi^\top N q}{\pi^\top N \mathbf{1}} \quad (47)$$

Integrating this rate over the horizon T_h , the probability of ruin is defined as the complement of the survival function:

$$P(\text{ruin}) = 1 - S(T_h) \leq 1 - \exp \left(-T_h \frac{\pi^\top N q}{\pi^\top N \mathbf{1}} \right) \quad (48)$$

Conclusion: The resulting bound captures the systemic interaction between regime duration (persistence), tail exposure (severity), and hazard structure (openness). The exponential term reflects the non-linear accumulation of risk, providing a robust measure of ruin probability in a non-ergodic environment. $\square$

Theorem A.7 (Tail-Sensitive Hazard Calibration). *Let ϵ_i denote the hazard rate for regime i . For a given baseline structural openness $\epsilon_{\min} > 0$ and a sensitivity constant $c > 0$, the hazard rate is calibrated as:*

$$\epsilon_i = \epsilon_{\min} + \frac{c}{\nu_i} \quad (49)$$

where ν_i represents the degrees of freedom (tail thickness) of the Student- t distribution associated with regime i . It follows that ϵ_i increases as ν_i decreases, indicating that regimes with heavier tails (smaller ν_i) exhibit higher systemic leakage risk.

Proof. The proof of the tail-sensitive mapping proceeds by analyzing the functional behavior of the calibration equation:

1. **Baseline Openness:** The condition $\epsilon_{\min} > 0$ ensures that the system remains an open dynamical system (substochastic) even in the limit of infinitely thin tails ($\nu_i \rightarrow \infty$).
2. **Sensitivity Control:** The constant $c > 0$ acts as a scaling factor that adjusts how aggressively the hazard rate responds to changes in tail thickness.
3. **Tail Thickness Analysis:** In a Student- t distribution, the parameter ν_i inversely correlates with tail heaviness; as $\nu_i \rightarrow \infty$, the distribution approaches Gaussian, while small ν_i values indicate extreme kurtosis and fat tails.
4. **Monotonicity:** Taking the partial derivative of ϵ_i with respect to ν_i :

$$\frac{\partial \epsilon_i}{\partial \nu_i} = -\frac{c}{\nu_i^2} \quad (50)$$

Since $c > 0$ and $\nu_i^2 > 0$, the derivative is strictly negative ($\frac{\partial \epsilon_i}{\partial \nu_i} < 0$).

Conclusion: As ν_i decreases (tails become heavier), the term $\frac{c}{\nu_i}$ increases, leading to a higher hazard rate ϵ_i . This formalizes the requirement that regimes characterized by extreme tail risk must exhibit higher systemic “leakage” or probability of ruin. $\square$

Theorem A.8 (Tail Risk Measurement Using Student- t Distribution). *Let X_t be a random variable representing asset returns at time t , and let $S_t \in \{1, \dots, K\}$ denote the latent market regime. Assume that for each regime i , the returns follow a Student- t distribution with location μ_i , scale σ_i , and degrees of freedom ν_i .*

1. **Regime-Specific Left-Tail Probability:** For a given significance level α and a corresponding empirical threshold x_α , the left-tail probability q_i for regime i is:

$$q_i = P(X_t \leq x_\alpha \mid S_t = i) = F_t(\dots \mid \mu_i, \sigma_i, \nu_i) \quad (51)$$

2. **Aggregate Tail Exposure ($\bar{q}$):** The aggregate systemic tail exposure is the residence-weighted average of the local tail probabilities:

$$\bar{q} = \sum_{i=1}^K \pi_i q_i \quad (52)$$

where π_i is the survival-weighted probability (residence weight) of regime i derived from the fundamental matrix N .

Proof. The measurement of aggregate tail risk follows from the conditional distribution properties of the Hidden Markov Model:

1. **Computation of q_i :** For each regime i , the probability mass residing below the threshold x_α is calculated using the Cumulative Distribution Function (CDF) of the fitted Student- t distribution:

$$q_i = \int_{-\infty}^{x_\alpha} \frac{\Gamma(\frac{\nu_i+1}{2})}{\sqrt{\nu_i\pi}\sigma_i\Gamma(\frac{\nu_i}{2})} \left(1 + \frac{1}{\nu_i} \left(\frac{x - \mu_i}{\sigma_i}\right)^2\right)^{-\frac{\nu_i+1}{2}} dx \quad (53)$$

This integral captures the heavy-tailed shocks unique to the volatility and kurtosis signatures of regime i .

2. **Integration of Systemic Exposure:** The aggregate exposure $\bar{q}$ is the expected value of the indicator function $I(X_t \leq x_\alpha)$ over the survival-weighted state space. By the law of total probability applied to the residence distribution π :

$$\bar{q} = \sum_{i=1}^K P(S_t = i)P(X_t \leq x_\alpha | S_t = i) = \sum_{i=1}^K \pi_i q_i \quad (54)$$

Substituting the Student- t CDF for q_i completes the proof.

□

Theorem A.9 (Structural Ruin Metrics). *The survival-weighted systemic risk of an open regime system is quantified by the following metrics:*

1. **Average Hazard Leakage ($\bar{\lambda}$):** *The integrated rate of probability mass dissipation across the survival geometry is computed as:*

$$\bar{\lambda} = \sum_{i=1}^K \pi_i \epsilon_i \quad (55)$$

where π_i is the residence weight and ϵ_i is the regime-specific hazard rate.

2. **Structural Ruin Probability Bound:** *Given a risk horizon T_h , the probability of the system crossing a ruin threshold is bounded by:*

$$P(\text{ruin}) \leq 1 - \exp\left(-T_h \frac{\pi^\top N q}{\pi^\top N \mathbf{1}}\right) \quad (56)$$

This metric integrates regime persistence (via the fundamental matrix N), tail severity (via the hazard vector q), and systemic openness (via π and ϵ) into a unified, survival-oriented risk measure. It formalizes the Talebian view of risk as a temporal survival problem rather than a static variance statistic.

B Integral Representation of the Fundamental Matrix

Proposition B.1. *If all eigenvalues of K have strictly negative real parts, then*

$$N = -K^{-1} = \int_0^\infty e^{Kt} dt.$$

Proof. Consider the matrix-valued function e^{Kt} . Since the eigenvalues of K have negative real parts, $e^{Kt} \rightarrow 0$ as $t \rightarrow \infty$. Also,

$$\frac{d}{dt} e^{Kt} = K e^{Kt}.$$

Integrating from 0 to T ,

$$\int_0^T K e^{Kt} dt = e^{KT} - I.$$

Premultiplying by K^{-1} ,

$$\int_0^T e^{Kt} dt = K^{-1}(e^{KT} - I).$$

Taking $T \rightarrow \infty$, and using $e^{KT} \rightarrow 0$,

$$\int_0^\infty e^{Kt} dt = -K^{-1}.$$

This is exactly N . □

C Proof of the Fragility Decomposition

Theorem C.1 (Fragility Decomposition). *Let π be a survival-weighted regime distribution, let $q = (q_1, \dots, q_K)^\top$ be the vector of regime-specific left-tail probabilities, and let ϵ be the hazard vector. Define*

$$F_{\text{cat}} = \pi^\top q, \quad F_{\text{haz}} = \pi^\top \epsilon.$$

Let F_{dur} be any positive scalar functional of the residence structure, for example

$$F_{\text{dur}} = \pi^\top \text{diag}(N).$$

Then the composite index

$$F = F_{\text{dur}} F_{\text{cat}} F_{\text{haz}}$$

is increasing in each component and therefore furnishes a multiplicative decomposition of structural fragility into residence, catastrophe, and hazard channels.

Proof. Assume all three components are strictly positive. Then

$$F(F_{\text{dur}}, F_{\text{cat}}, F_{\text{haz}}) = F_{\text{dur}} F_{\text{cat}} F_{\text{haz}}.$$

The partial derivatives are

$$\frac{\partial F}{\partial F_{\text{res}}} = F_{\text{cat}} F_{\text{haz}} > 0,$$

$$\frac{\partial F}{\partial F_{\text{cat}}} = F_{\text{dur}} F_{\text{haz}} > 0,$$

$$\frac{\partial F}{\partial F_{\text{haz}}} = F_{\text{dur}} F_{\text{cat}} > 0.$$

Hence F is strictly increasing in each argument. Therefore an increase in residence mass, catastrophe severity, or hazard leakage increases total fragility. $\square$

D Spectral Fragility Principle

Theorem D.1 (Spectral Fragility Principle). *Suppose $Q(\theta)$ is a family of substochastic matrices depending smoothly on a scalar parameter θ , with $\rho(Q(\theta)) < 1$ for all admissible θ . Let*

$$N(\theta) = (I - Q(\theta))^{-1}.$$

If $Q(\theta)$ increases entrywise in θ , then $N(\theta)$ increases entrywise in θ , and in particular $\rho(N(\theta))$ is nondecreasing in θ .

Proof. Because $\rho(Q(\theta)) < 1$, we have the convergent Neumann expansion

$$N(\theta) = \sum_{m=0}^{\infty} Q(\theta)^m.$$

If $Q(\theta)$ increases entrywise in θ , then for every positive integer m , the matrix power $Q(\theta)^m$ also increases entrywise in θ , since matrix multiplication preserves order for nonnegative matrices. Hence each term in the series is nondecreasing entrywise, and therefore the sum $N(\theta)$ is nondecreasing entrywise. Now $N(\theta)$ is nonnegative. By Perron–Frobenius theory, the spectral radius of a nonnegative matrix is monotone with respect to entrywise order. Therefore $\rho(N(\theta))$ is nondecreasing in θ . $\square$

E Residence-Weighted Ruin Bound

Theorem E.1 (Residence-Weighted Ruin Bound). *Let $q = (q_1, \dots, q_K)^\top$ be the vector of regime-specific tail-loss probabilities, let π be a survival-weighted regime distribution, and let N be the fundamental matrix. Define the effective ruin intensity*

$$\Lambda = \frac{\pi^\top N q}{\pi^\top N \mathbf{1}}.$$

Then over a horizon $T_h > 0$, a structural ruin bound is given by

$$P(\text{ruin}) \leq 1 - e^{-T_h \Lambda}.$$

Proof. Interpret Λ as the average tail-event intensity under residence weighting. The numerator $\pi^\top N q$ aggregates regime-specific tail probabilities weighted by expected survival-adjusted time spent in each state, while the denominator $\pi^\top N \mathbf{1}$ normalizes by total expected residence time. Thus Λ is an average rate of ruin-relevant exposure.

If ruin events are dominated by a hazard process with effective rate at most Λ , then the survival probability over horizon T_h is bounded below by the exponential survival law,

$$P(\text{survival over } [0, T_h]) \geq e^{-T_h \Lambda}.$$

Therefore,

$$P(\text{ruin}) = 1 - P(\text{survival}) \leq 1 - e^{-T_h \Lambda}.$$

This yields the stated bound. □

F Existence of the Fundamental Matrix

Proposition F.1. *If Q is substochastic and transient, then $I - Q$ is invertible and*

$$N = (I - Q)^{-1} = \sum_{m=0}^{\infty} Q^m.$$

Proof. Transience implies $\rho(Q) < 1$. By a standard result in matrix analysis, if the spectral radius of Q is strictly less than one, then the Neumann series converges absolutely:

$$\sum_{m=0}^{\infty} Q^m.$$

Moreover,

$$(I - Q) \sum_{m=0}^{\infty} Q^m = \sum_{m=0}^{\infty} Q^m - \sum_{m=1}^{\infty} Q^m = I.$$

Similarly,

$$\left(\sum_{m=0}^{\infty} Q^m \right) (I - Q) = I.$$

Hence $(I - Q)^{-1}$ exists and equals the convergent series. □

G Interpretive Corollary: Duration versus Catastrophe Fragility

Corollary G.1. *Two systems may have different total fragility rankings depending on whether the ranking criterion emphasizes residence structure or tail severity. In particular:*

1. *a system with larger F_{dur} but smaller F_{cat} is more duration-fragile;*
2. *a system with smaller F_{dur} but larger F_{cat} is more catastrophe-fragile.*

Proof. This follows immediately from the multiplicative decomposition

$$F = F_{\text{dur}} F_{\text{cat}} F_{\text{haz}}.$$

Holding F_{haz} fixed, the ranking depends on the relative sizes of the residence and catastrophe components. A system can dominate on one margin and be dominated on another. Hence fragility is not one-dimensional. □

Masthead Graphic

The mast head collage features the famous Jugendstil (Art Nouveau) architectural elements found on Alberta iela (Albert Street) in Rīga, Latvia. These specific reliefs and sculptures are classic examples of the “Eclectic” or “Decorative” Art Nouveau style designed by architect Mikhail Eisenstein in the early 1900s. The top left shows a rounded, semicircular stone balcony jutting out from a building corner under a clear blue sky. Above the central window frame sits a sculpted stone mascarón (a stylized human face) expressing an intense, open-mouthed expression, flanked by heavy decorative brackets and rusticated pilasters along the wall. The top right image displays a winged lion relief. It depicts a powerful winged lion rearing upward, sculpted in profile but facing forward with an open, roaring mouth. The mane and wings are rendered with sweeping, fluid curvilinear lines typical of Art Nouveau ornamentation, set against a deeply grooved, horizontal rusticated wall background. The bottom left winged mascarón is a calm, idealized female face adorned with a stylized headdress. Stretching horizontally from each side of her head are large, highly detailed outspread stone wings that anchor the relief to the surrounding architectural frame. The bottom center features a free-standing Egyptian Sphinx sculpture flanking an entranceway along Alberta iela. The sculpture depicts a winged sphinx featuring a female bust, geometric Egyptian-style headdress, and large curled wings along its flanks. It rests in a couchant (lying down) profile position on a heavy stone plinth, echoing the geometric and symbolic architectural trends of the late Art Nouveau period. Finally, the bottom right image features a floral vault and medallion. The underside is painted a warm terracotta/salmon color, decorated with intricate white stucco floral and leaf motifs. Directly underneath, serving as a decorative keystone between two arched windows, is a beautifully carved female head emerging from fluid, root-like organic stone carvings.

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